

Due on monday November 7th, at the beginning of tutorial.

[**SELF**] means: will not be marked. Working them out may be helpful for doing the other problems.

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### 1. Dirac notation (bra-ket notation).

Work through an introduction to Dirac notation (Nash notes Sec. 2.3, wikipedia page, a q.m. textbook,...) Some links are on the module webpage.

The bra vector  $\langle\phi|$  is sometimes called the dual vector or co-vector of the ket vector  $|\phi\rangle$ .

- (a) [**1 pt.**] If  $|\phi\rangle = c|\chi\rangle$ , where  $c$  is a complex number, express  $\langle\phi|$  in terms of  $\langle\chi|$ .
- (b) [**1 pt.**] If  $|\phi\rangle = \hat{A}|\chi\rangle$ , where  $\hat{A}$  is an operator (not necessarily hermitian), express  $\langle\phi|$  in terms of  $\hat{A}$  and  $\langle\chi|$ .
- (c) [**SELF**] Is  $\langle\chi|\phi\rangle$  a number, an operator, or a ket vector, or a bra vector? What about  $|\chi\rangle\langle\phi|$ ?
- (d) [**1 pt.**] If  $|\phi\rangle$  is not normalized, what would you multiply it by, in order to obtain a normalized state vector? Are you multiplying by a number, an operator, or another ket?
- (e) [**1 pt.**] If  $|\chi\rangle$  is normalized and  $|\phi\rangle = \gamma|\chi\rangle$ , where  $\gamma$  is a complex number, find the value of  $\langle\chi|\phi\rangle$ .
- (f) [**1 pt.**] If  $\langle\chi|\phi\rangle = \beta$ , then what is  $\langle\phi|\chi\rangle$ ?

### 2. Basis vectors.

A physical system is described by a three-dimensional Hilbert space. Any state vector can be expressed as a linear combination of the three state vectors  $|\phi_1\rangle$ ,  $|\phi_2\rangle$ ,  $|\phi_3\rangle$ . We say that the set  $\{|\phi_j\rangle\}$  is a basis set (a set of basis vectors) for this space.

We choose our three basis vectors to be **orthonormal**.

- (a) [**1 pt.**] Find out (look up) what orthonormal means. Our basis set  $\{|\phi_j\rangle\}$  is orthonormal: Express what this means in terms of inner products. In particular, what is  $\langle\phi_1|\phi_1\rangle$ , and what is  $\langle\phi_1|\phi_2\rangle$ ?

- (b) [2 pts.] Consider the normalized state

$$|W\rangle = \gamma |\phi_1\rangle + \gamma |\phi_2\rangle + \gamma |\phi_3\rangle.$$

Determine  $\gamma$ . Assume  $\gamma$  to be purely imaginary.

- (c) [2 pts.] Consider the states

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{3}} |\phi_1\rangle - \frac{1}{\sqrt{3}} |\phi_2\rangle + \frac{i}{\sqrt{3}} |\phi_3\rangle \\ |\theta\rangle &= \frac{i}{2} |\phi_1\rangle + \frac{\sqrt{3}}{2} |\phi_3\rangle \end{aligned}$$

Show that these two states are normalized.

- (d) [1 pt.] Express  $\langle\psi|$  in terms of the bra vectors corresponding to the basis vectors, i.e., in terms of the set  $\{\langle\phi_j|\}$ .
- (e) [2 pts.] Calculate  $\langle\psi|\theta\rangle$  and  $\langle\theta|\psi\rangle$ .
- (f) [3 pts.] Let's define the operator  $\hat{A} = |\theta\rangle\langle\psi|$ . Write out the operator in terms of the basis vectors and their duals, i.e., in terms of operators  $|\phi_i\rangle\langle\phi_j|$ .
- (g) [1 pt.] Find the state obtained by operating with  $\hat{A}$  on the state  $|\phi_1\rangle$ .
- (h) [2 pts.] Find the state obtained by operating with  $\hat{A}$  on the state  $|W\rangle$ , i.e., express  $\hat{A}|W\rangle$  in terms of the basis vectors.
- (i) [2 pts.] Let's introduce the shorthand notation  $A_{ij} = \langle\phi_i|\hat{A}|\phi_j\rangle$ . Find the values of the nine quantities  $A_{ij}$ . These are generally known as 'matrix elements' of the operator  $\hat{A}$ .

### 3. Vector/matrix notation. (Building on previous problem)

We are using the set  $\{|\phi_j\rangle\}$  as a basis set. This can be expressed in vector notation as

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |\phi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

The size of the vectors is the number of orthonormal basis vectors required to represent all states, which is also the dimension of the Hilbert space. (In this case: three.)

- (a) [1 pt.] Write down the states  $|W\rangle$ ,  $|\psi\rangle$  and  $|\theta\rangle$  (from previous problem) as vectors.
- (b) [2 pts.] Write down the row vectors which represent the dual (bra) vectors  $\langle\psi|$  and  $\langle\theta|$ .
- (c) [1 pt.] Using these representations, calculate  $\langle\psi|\theta\rangle$  through matrix multiplication between a row vector and a column vector. (Check with the value that you calculated previously using bra-ket notation.)
- (d) [3 pts.] Construct the matrix representing the operator  $\hat{A} = |\theta\rangle\langle\psi|$ , by multiplying the column vector representing  $|\theta\rangle$  and the row vector representing  $\langle\psi|$ . (The order of vectors is important!)
- (e) [SELF] Is it clear why  $A_{ij} = \langle\phi_i|\hat{A}|\phi_j\rangle$  are called the “matrix elements”?
- (f) [2 pts.] Using matrix-vector multiplication, find the state obtained by operating with  $\hat{A}$  on the state  $|W\rangle$ .

4. A particle in a 1D harmonic oscillator: eigenstates.

For a single particle in a harmonic oscillator potential  $V(x) = \frac{1}{2}m\omega^2 x^2$ , the time-independent Schroedinger equation has an infinite number of eigenvalue/eigenfunction pairs. The lowest three energy eigenvalues are  $E_0 = \frac{1}{2}\hbar\omega$ ,  $E_1 = \frac{3}{2}\hbar\omega$ ,  $E_2 = \frac{5}{2}\hbar\omega$ . The corresponding eigenfunctions are respectively

$$\begin{aligned}\phi_0(x) &= A_0 e^{-x^2/2\sigma^2}, & \phi_1(x) &= A_1 x e^{-x^2/2\sigma^2}, \\ \phi_2(x) &= A_2 \left( \frac{2x^2}{\sigma^2} - 1 \right) e^{-x^2/2\sigma^2}. & \text{Here } \sigma^2 &= \frac{\hbar}{m\omega}.\end{aligned}$$

- (a) [1+1 pts.] Sketch (by hand) plots of the three functions  $\phi_n(x)$  for  $n = 0, n = 1, n = 2$ . You may use a computer program if you want to figure out how these functions look like, but please submit hand-drawn sketches.

By looking at the three plots and observing the pattern, guess and sketch the plots of the next two eigenstates,  $\phi_3(x)$  and  $\phi_4(x)$ .

- (b) [2 pts.] Look up (or calculate, or ask a computer algebra system or online integrator) the integrals

$$\int_{-\infty}^{+\infty} x^{2n} e^{-x^2} dx$$

for  $n = 0$ ,  $n = 1$ ,  $n = 2$ . Report the three results.

The first one ( $n = 0$ , integral over a Gaussian) is important enough that you might consider learning it for life.

- (c) [**SELF**] Calculate the integrals

$$\int_{-\infty}^{+\infty} x^{2n} \exp \left[ -\frac{x^2}{2\sigma^2} \right] dx$$

for  $n = 0$ ,  $n = 1$ ,  $n = 2$ , based on the integrals above.

- (d) [**SELF**] Use symmetry arguments to evaluate the integral

$$\int_{-\infty}^{+\infty} x^{2n+1} \exp \left[ -\frac{x^2}{2\sigma^2} \right] dx$$

for any integer  $n$ . The integrand is an odd function.

- (e) [**1+1+2 pts.**] Show that  $\phi_0$  and  $\phi_1$  are orthogonal to each other, as are  $\phi_1$  and  $\phi_2$ , and finally that  $\phi_0$  and  $\phi_2$  are orthogonal to each other.
- (f) [**5 pts.**] Show in general that, if  $|w_1\rangle$  and  $|w_2\rangle$  are eigenstates of a hermitian operator corresponding to distinct eigenvalues, then they are orthogonal to each other. (This is an important result.)
- (g) [**1+2 pts.**] Calculate  $A_0$  and  $A_1$  so that  $\phi_0$  and  $\phi_1$  are normalized.
- (h) [**SELF**] Calculate  $A_2$  so that  $\phi_2$  is normalized.
- (i) [**1 pt.**] Construct a solution of the TDSE, based on the three eigenfunctions of the time-independent SE.
- (j) [**3 pts.**] At time  $t = 0$ , the particle in the harmonic oscillator finds itself in the state

$$\psi(x, 0) = \frac{1}{\sqrt{2}}\phi_0(x) - \frac{1}{2}\phi_1(x) + \frac{1}{2}\phi_2(x).$$

What is the wavefunction (state)  $\psi(x, t)$  at a later time  $t$ ?

- (k) [**SELF**] If the  $\phi_n(x)$  functions are normalized, then show that  $\psi(x, 0)$  is normalized, and that  $\psi(x, t)$  remains normalized at later times  $t$ .
- (l) This is a good time to have a first browse through Nash Chapter 4 (The harmonic oscillator), or the wikipedia page on “Quantum Harmonic Oscillator”.